A HYBRID OPTIMIZING AND CLUSTERING LUNG NODULE MALIGNANCY USING MULTI-PROTOTYPE ATTENTION CAPSULE FRAMEWORK

21CSP401L -MAJOR PROJECT REPORT

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Certified that this project report titled **"A HYBRID OPTIMIZING AND CLUSTERING LUNG NODULE MALIGNANCY USING MULTI-PROTOTYPE ATTENTION CAPSULE FRAMEWORK"** is the Bonafide work of **T. SRAVAN KUMAR [REG NO: RA2211028020043], B. CHARAN REDDY [REG NO: RA2211028020034], K. MOKSHAK SRI [REG NO: RA2211028020029], P. HIMAWANTH REDDY [REG NO: RA2211028020016]** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form any other project report or dissertation on the basis of which a degree or award was conferred on an occasion on this or any other candidate.

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INTERNAL EXAMINER I

INTERNAL EXAMINER II

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DECLARATION

We hereby declare that the entire work contained in this project report titled **"A HYBRID OPTIMIZING AND CLUSTERING LUNG NODULE MALIGNANCY USING MULTI-PROTOTYPE ATTENTION CAPSULE FRAMEWORK"** has been carried out by **T. SRAVAN KUMAR [REG NO: RA2211028020043]**, **B. CHARAN REDDY [REG NO: RA2211028020034]**, **K. MOKSHAK SRI [REG NO: RA2211028020029]**, **P. HIMAWANTH REDDY [REG NO: RA2211028020016]** at SRM Institute of Science and Technology, Ramapuram, Chennai- 600089, under the guidance of **Ms. J. Arthy M.E.,(Ph.D.), Assistant Professor,** Department of Computer Science and Engineering W/S in Big Data Analytics and Cloud Computing.

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Own Work Declaration form

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ABSTRACT

Deep learning techniques have become fundamental in computer-assisted diagnosis, particularly in thoracic computed tomography analysis for lung cancer detection. Lung nodules exhibit complex morphological characteristics such as texture, spiculation, calcification, and internal structure, which significantly influence malignancy assessment. Furthermore, prototype learning enables interpretable decision-making by associating embeddings with representative class centroids. The base study introduced an efficient capsule-prototype hybrid that predicts lung nodule visual attributes and malignancy scores using clustering-driven supervision on the LIDC-IDRI dataset. It achieved competitive multiclass malignancy prediction while drastically reducing backbone parameters compared to classical capsule architectures, thereby improving computational feasibility and interpretability. However, several limitations remain in the base framework. First, clustering quality measured through a modified Davies-Bouldin Index remains moderate, indicating imperfect separation between attribute-specific clusters. Second, imbalance in lung nodule visual attribute distributions affects stability and representation learning, particularly for attributes with dominant class frequencies. Third, certain high-level capsules converge prematurely due to imbalance between attribute and malignancy loss terms, resulting in disproportionate influence on malignancy prediction. Fourth, reliance on 2D slice representations may restrict contextual feature extraction from volumetric CT scans. To overcome these drawbacks, we propose a Hybrid Multi-Prototype Attention Capsule Framework integrating Efficient Capsule Network with Adaptive Multi-Centroid Contrastive Clustering and Distribution-Aware Optimization. The proposed system combines Self-Attention Routing Mechanism, Adaptive Weighted Davies-Bouldin Multi-Centroid Loss, Contrastive Prototype Regularization, and Distribution Alignment using Kull back-Leibler Divergence. Compared with the base Efficient Capsule Network and classical Dynamic Routing Capsule Network, our hybrid model introduces Hierarchical Attention-Prototype Alignment and Balanced Multi- Stage Optimization. A dual-stage training strategy first optimizes Lung Nodule Visual Attribute capsules independently using Adaptive Cluster Compactness Enhancement, followed by malignancy distribution refinement using Distribution Consistency Regularization. This hybrid approach enhances cluster separation, stabilizes capsule specialization, improves multiclass malignancy prediction Robustness, and maintains parameter efficiency while significantly improving interpretability.

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LIST OF ACRONYMS AND ABBREVIATIONS

CNN	CONVOLUTIONAL NEURAL NETWORK
HLC	HIGH-LEVEL CAPSULE
LNVA	LUNG NODULE VISUAL ATTRIBUTE
LNM	LUNG NODULE MALIGNANCY
DBI	DAVIES-BOULDIN INDEX
LDBI	DAVIES-BOULDIN INDEX LOSS
LLNVA	LUNG NODULE VISUAL ATTRIBUTE LOSS
LLNM	LUNG NODULE MALIGNANCY LOSS
LREC	IMAGE RECONSTRUCTION LOSS
KL	KULLBACK-LEIBLER DIVERGENCE
ROI	REGION OF INTEREST
DSC	DEPTHWISE SEPARABLE CONVOLUTION
SGD	STOCHASTIC GRADIENT DESCENT
CAD	COMPUTER ASSISTED DIAGNOSIS
CT	COMPUTER TOMOGRAPHY

CHAPTER 1

INTRODUCTION

Lung cancer remains one of the leading causes of cancer-related mortality worldwide, making early detection and accurate diagnosis critical for improving patient survival rates. Computer-assisted diagnosis (CAD) systems, particularly those analyzing thoracic computed tomography (CT) scans, have become fundamental tools for radiologists. However, assessing the malignancy of lung nodules is highly complex due to their heterogeneous morphological characteristics, which include variations in texture, spiculation, calcification, and internal structure.

While deep learning techniques, specifically Convolutional Neural Networks (CNNs), have driven significant advancements in medical image analysis, they inherently struggle to model spatial hierarchies and part-whole relationships. To address this, Capsule Networks have emerged as a powerful alternative. By encoding pose-aware representations, capsule networks can better capture the complex spatial orientations of lung nodules. When combined with prototype learning, these networks also offer a highly desirable trait in the medical field: interpretability, as decision-making can be associated with representative class centroids (prototypes).

1.1 PROBLEM STATEMENT

Accurate and interpretable lung nodule malignancy prediction remains a significant challenge in medical image analysis. While capsule networks provide structured representation learning, existing prototype-based capsule models exhibit moderate clustering separation, imbalance sensitivity, and instability between attribute and malignancy optimization stages. The interaction between clustering compactness, inter-cluster separation, and malignancy distribution alignment is not fully optimized, resulting in suboptimal latent space organization. Moreover, high-level capsules may encode overlapping attribute representations, thereby reducing transparency in malignancy reasoning pathways.

Mathematically, the clustering objective can be expressed as minimizing intra-cluster dispersion while maximizing inter-cluster separation:

where S represents intra-cluster compactness and d represents inter-cluster centroid separation. However, single-stage optimization does not ensure balanced capsule specialization and malignancy distribution consistency. Therefore, an enhanced hybrid framework is required to optimize clustering stability, interpretability, and predictive robustness simultaneously.

1.2 AIM OF THE PROJECT

The primary aim of this project is to design and develop a Hybrid Multi-Prototype Attention Capsule Framework for the accurate, interpretable, and parameter-efficient evaluation of lung nodule malignancy in thoracic CT scans. Specifically, the project seeks to overcome the limitations of traditional deep learning models and existing baseline capsule networks by achieving the following targeted objectives:

1. Enhance Cluster Compactness

To design an adaptive multi-centroid clustering mechanism that minimizes intra-cluster variance while maximizing inter-cluster separation in high-level capsule latent space.

2. Stabilize Capsule Specialization

To implement multi-stage optimization that prevents premature convergence of specific attribute capsules and ensures balanced learning across all lung nodule visual attributes.

3. Improve Malignancy Distribution Modeling

To incorporate distribution-aware Kull back-Leibler Divergence alignment for better modeling of inter-radiologist variability in malignancy annotations.

4. Maintain Parameter Efficiency

To preserve lightweight backbone architecture while integrating additional attention and contrastive regularization modules.

5. Increase Interpretability Transparency

To strengthen prototype-to-capsule alignment mechanisms that provide clinically meaningful reasoning consistent with standardized diagnostic guidelines.

1.3 PROJECT DOMAIN

This project operates at the intersection of Artificial Intelligence and Healthcare Informatics, specifically falling within the domains of Deep Learning and Medical Image Analysis. At its core, the research leverages advanced computational methodologies—namely Capsule Networks, Self-Attention Mechanisms, and contrastive clustering algorithms—to enhance Computer-Assisted Diagnosis (CAD) systems. By applying these highly parameter-efficient neural architectures to thoracic CT(computed tomography) scans, the project bridges the gap between raw computational modeling and practical clinical applications, ultimately providing a robust and interpretable tool for oncology and lung cancer detection. Despite its advantages, lung nodule characterization using semantic segmentation faces several challenges. Variability in CT image quality, differences in scanner settings, and presence of noise can affect model performance. Small nodules are particularly difficult to detect due to their size and similarity to surrounding tissues. Limited availability of labelled medical data also poses a challenge for training deep learning models. Ensuring model interpretability is important for clinical acceptance, as doctors need to understand the basis of predictions. Researchers are exploring advanced techniques such as transfer learning, attention mechanisms, and explainable AI to address these challenges. As artificial intelligence continues to evolve, lung segmentation systems are expected to become more accurate, reliable, and widely used in medical practice.

1.4 SCOPE OF PROJECT

This project focuses on building and testing a specialized deep learning model designed to determine whether lung nodules in thoracic CT scans are malignant or benign. Rather than addressing general respiratory diseases, the work is strictly limited to analyzing specific physical traits of lung nodules, such as their texture, spiculation, calcification, and internal structure, using the LIDC-IDRI dataset.

On the technical side, the core of the project involves developing the Hybrid Multi-Prototype Attention Capsule Framework. A major part of this work is replacing older, traditional dynamic routing methods with a new Self-Attention Routing Mechanism. This shift is intended to help the network process spatial features much more efficiently.

The scope also covers the implementation of a unique two-step training process. First, the model will be trained to tightly group the visual attributes of the nodules. Following that, techniques like Kull back-Leibler (KL) Divergence and Contrastive Prototype Regularization will be used to refine how the system actually predicts the malignancy.

Additionally, a major boundary of this research is avoiding the "black-box" problem common in AI. We want the decision-making process to be transparent. Therefore, a significant portion of the project is dedicated to making the model's internal groupings clear, so that the final clinical predictions are highly interpretable. Ultimately, the project aims to solve the known issues in current Efficient Capsule Networks—such as poor data clustering and class imbalance—without making the model too computationally heavy for practical clinical use.

1.5 MOTIVATION

Deep learning models in healthcare must achieve not only high predictive accuracy but also transparent reasoning. Black-box malignancy classifiers lack structured explanations aligned with clinical guidelines, limiting physician trust and regulatory acceptance. Although capsule networks introduce structural modeling capabilities, their integration with prototype learning still requires improved cluster stability and balanced loss optimization. Addressing these challenges is critical for advancing trustworthy artificial intelligence in medical diagnosis. Furthermore, improving latent space organization through adaptive multi-centroid clustering and attention-based prototype refinement can significantly enhance interpretability and generalization. Lightweight yet structured architectures enable deployment in real-world clinical environments with limited computational resources. This motivates us to do this project.

1.6 METHODOLOGY

The proposed system introduces a Hybrid Multi-Prototype Attention Capsule Framework designed specifically to evaluate lung nodule malignancy in thoracic CT scans. To overcome the computational inefficiencies of traditional models, the architecture replaces standard dynamic routing with a Self-Attention Routing Mechanism, allowing the network to efficiently extract complex spatial features and part-whole relationships. This is combined with Adaptive Multi-Centroid Contrastive Clustering, which dynamically groups distinct visual characteristics of the nodules rather than relying on rigid, pre-set thresholds. Finally, the framework employs a Balanced Multi-Stage Optimization strategy that first trains the model to understand visual attributes (like texture and spiculation) before refining the actual malignancy diagnosis. This dual-stage approach ensures the model remains parameter-efficient while providing transparent, interpretable reasoning for its clinical predictions.

CHAPTER 2

LITERATURE SURVEY

In Future years there has been a lot of passion in integration of deep learning techniques into systems that anticipate lung disease and offers therapeutic solutions and overview of major studies that have best the applications of learning in this field is given in the review of literature

[1] M. Rodrigues, Hélder P. Oliveira, Tania Pereira Efficient-Proto-Caps: A Parameter-Efficient and Interpretable Capsule Network for Lung Nodule Characterization. Efficient-Proto-Caps addresses the interpretability gap in medical AI by offering a lightweight model that maintains high accuracy with only 2% of the baseline's parameters.

[2] Yohei, Yutaro Shigeto - Characterization of Pulmonary Nodules in Computed Tomography Images Based on Pseudo-Labelling Using Radiology Reports. This research demonstrates that training CAD systems using pseudo-labels automatically extracted from radiology reports achieves diagnostic accuracy comparable to expert-annotated models, effectively overcoming the bottleneck of requiring extensive medical image labelling

[3] S. Malarvannan- a review of deep learning techniques used for the classification of lung cancer BI deep learning techniques particularly cities the classifications under segmentation provide a potent and effective way to automate lung cancer detection greatly enhancing the scalability and precision of the manual diagnosis

[4] Rodina, Adel Mohamed - This study developed an automated system to diagnose Pneumothorax in neonates by using deep learning to extract features from lung ultrasounds. By employing an CNN model to automatically generate M-modes and detect "lung sliding," the system bypassed time-consuming manual evaluations and achieved 100% classification accuracy on test cases, offering a fast and highly efficient diagnostic tool for clinical use.

[5] Maria Ghita -this paper aims to analyse the contribution and application of oscillation technique cube devices in lung cancer assessment two devices and corresponding methods can be feasible to distinguish among different degrees of lungs tissue heterogeneity.

[6] Surya Majumder, - Addressing the formidable challenges posed by the diagnosis and management of various types of cancer, including breast, colon, lung, and colorectal cancer, demands innovative solutions to streamline histopathological analysis processes.

[7] Mattakoyya Aharonu- This paper introduces a dual-pipeline deep learning framework featuring LCSC Net for automated lung cancer subtype classification and LCSA Net for patient survival analysis. The proposed models achieved over 95% accuracy on standard datasets, outperforming existing methods and demonstrating strong potential for real-world automated healthcare diagnostics.

[8] Richard Lupat, - This research introduces Moanna, a novel deep-learning algorithm that accurately predicts breast cancer subtypes by integrating complex multi-omics data through a semi-supervised Autoencoder and multi-task learning. Evaluated on independent datasets, the model achieved high classification accuracy (up to 98%) and successfully identified cancer subtypes that correlate significantly better with patient survival than traditional methods

[9] Kirsten Taphorn - demonstrates that combining grating-based X-ray dark-field imaging with spectral imaging can effectively diagnose and differentiate structural lung diseases, such as fibrosis and emphysema, in human chest radiography. By overcoming the high-radiation and low-sensitivity limitations of traditional methods, this novel approach offers a highly sensitive diagnostic tool at a safe, clinically relevant radiation dose.

CHAPTER 3

PROJECT DESCRIPTION

3.1 EXISTING SYSTEM

The existing system, named Efficient-Proto-Caps, presents a novel deep learning architecture designed for the characterization of lung nodules from computed tomography (CT) images, with a dual focus on maintaining high diagnostic accuracy while ensuring parameter efficiency and inherent interpretability. The core problem this system addresses is a significant barrier to the adoption of AI in clinical practice: the 'black box' nature of many high-performing deep learning models. While models often achieve or exceed human-level performance in tasks like malignancy prediction, their decision-making processes are opaque, making it difficult for clinicians to trust or verify their outputs. This lack of transparency is particularly problematic in high-stakes medical diagnoses where understanding the rationale behind a prediction is crucial for patient care. The existing methodology aims to bridge this gap by creating a model that not only predicts the likelihood of malignancy for a given lung nodule but also explains its prediction by characterizing the nodule based on a set of visual attributes that are directly aligned with established clinical guidelines, such as those from the Fleischner Society and the American College of Radiology.

The system's purpose is to function as a computer-assisted diagnostic tool that provides radiologists with a transparent, evidence-based second opinion, thereby increasing diagnostic confidence and workflow efficiency. The scope of the work involves developing a lightweight model that can be deployed in resource-constrained environments without sacrificing the nuanced, multi-faceted analysis required for accurate lung nodule assessment. The system is specifically engineered to be interpretable by design, rather than relying on post-hoc explanation methods, which are often unreliable and may not faithfully represent the model's true internal logic. This inherent interpretability is achieved by synergistically combining the hierarchical feature-learning capabilities of capsule networks with the case-based reasoning paradigm of prototype learning. The primary application area for the Efficient-Proto-Caps system is in the field of diagnostic radiology, specifically for lung cancer screening and diagnosis programs. Within this domain, it serves as a sophisticated decision support system for radiologists tasked with interpreting large volumes of thoracic CT scans. The system's ability to provide both a malignancy score and an analysis of underlying visual attributes—such as texture, spiculation, and margin—makes it a powerful tool for standardizing nodule evaluation and reducing inter-observer variability, which is a common challenge in radiology. The potential industry impact extends a0cross the healthcare and clinical AI sectors. By offering a model that is both accurate and transparent, it can accelerate the regulatory approval and clinical integration of AI-powered diagnostic tools. Its parameter-efficient design, using a backbone with only about 2% of the parameters of its baseline model, makes it economically viable for wider deployment in hospitals and imaging centers that may not have access to high-end GPU hardware. Beyond the immediate field of medical imaging, the fundamental approach of the existing system has broader applicability in any industry where final classifications are grounded in concept-based or example-based attributes. For instance, in manufacturing and quality control (Industry 4.0), it could be used for visual anomaly detection, where the model would not only flag a defect but also classify the type of defect based on learned visual prototypes. In finance and FinTech, a similar architecture could be applied to fraud detection, where the model explains its reasoning by pointing to specific transaction patterns that resemble known fraudulent prototypes.

The system's design principles are also relevant for environmental and sustainability applications, such as satellite imagery analysis for monitoring deforestation or agricultural land use, where classifications can be explained through intermediate, concept-level features. The methodology of the Efficient-Proto-Caps system is a carefully constructed synthesis of several advanced deep learning concepts. The model architecture begins with a backbone based on Efficient-CapsNet, chosen for its computational efficiency and robust representation learning. This backbone, which includes convolutional layers, residual blocks, and a squeeze-and-excitation block, processes input 2D image patches of lung nodules and generates a set of low-level feature embeddings. These features are then organized into low-level capsules. A key innovation of the backbone is the use of a self-attention routing mechanism, which replaces the computationally expensive dynamic routing of traditional capsule networks. This mechanism routes the low-level capsules to form eight high-level capsules, each designed to instantiate a specific lung nodule visual attribute (LNVA), such as sphericity, lobulation, or texture. The core of the model's interpretability is the prototype layer. This layer contains a set of learnable prototypes for each possible score of each LNVA. During training, the high-level capsules are encouraged to cluster around the prototypes corresponding to their ground-truth LNVA score. This clustering is enforced by a novel, differentiable loss function based on the Davies-Bouldin Index (DBI), which was modified to accommodate multiple centroids (prototypes) per cluster. This loss function, L_DBI, promotes high intra-cluster compactness and high inter-cluster separability in the latent space. Additional loss functions guide the training, including a mean squared error loss (L_LNVA) on the predicted LNVA scores and a Kullback-Leibler divergence loss (L_LNM) that aligns the model's predicted malignancy distribution with the distribution derived from multiple radiologists' scores. This use of a distribution rather than a single mean value allows the model to learn and represent the inherent uncertainty and inter-observer variability in the diagnostic task. Finally, a reconstruction loss (L_Rec) acts as a regularizer, ensuring the high- level capsules retain sufficient information to reconstruct a pseudo-segmentation of the input nodule.

The existing system was rigorously validated using the Lung Image Database Consortium image collection (LIDC-IDRI), which is widely recognized as the standard benchmark for this task. The validation process involved a comprehensive data preprocessing pipeline to normalize the heterogeneous CT scan data, including Hounsfield unit clipping, min-max normalization, and resampling to uniform slice thickness and pixel spacing. Lung nodules rated as indeterminate (malignancy score of 3) or those annotated by fewer than three radiologists were excluded to ensure a high-quality dataset for training and testing. The experimental setup employed a 5-fold cross-validation strategy to ensure the robustness and generalizability of the results, with the experiment repeated three times with different random seeds. The model's performance was primarily assessed using the 'within-1' accuracy metric, which considers a prediction correct if it is within ±1 of the ground-truth integer score, a common practice for ordinal classification tasks in medical imaging to account for minor disagreements in scoring. The key result highlighted is an overall 'within-1' accuracy of 89.7% for the combined task of predicting lung nodule malignancy and all associated visual attributes. This performance was found to be statistically comparable to the baseline Proto-Caps model. For the specific task of malignancy prediction, the system achieved a within-1 accuracy of 87.7%, an AUC of 0.94, and an F1-score of 0.88. The evaluation also included an assessment of the clustering quality in the latent space, reporting an average DBI of 0.95 across all LNVAs, indicating a moderate to high quality of cluster formation, which is essential for the prototype- based explanation mechanism.

The primary strength of the Efficient-Proto-Caps system is its successful integration of high performance, computational efficiency, and inherent interpretability into a single, cohesive framework. By providing a decision rationale that is aligned with globally accepted clinical guidelines, it represents a significant step towards developing more transparent and clinically viable computer-assisted diagnostic systems. The use of a backbone with only a fraction of the parameters of its baseline demonstrates that complex, 'black box' models are not always necessary to achieve strong performance, opening the door for deployment on less powerful hardware. The novel modification of the Davies-Bouldin Index as a loss function is another key strength, providing an effective mechanism for enforcing meaningful structure in the model's latent space. However, the system is not without limitations. Its performance, while strong, was outperformed by some state-of-the-art models that focus solely on binary malignancy classification without providing interpretability. The ablation studies showed that the primary experimental design was optimal, but also revealed that prediction accuracy for certain attributes like sphericity and lobulation was lower than the baseline. A notable limitation discovered during a post-hoc analysis was that the high-level capsule for 'internal structure' inadvertently encoded information more relevant to malignancy prediction than the attribute itself, due to the low variability of this feature in the dataset. This suggests that the disentanglement of features is not yet perfect and could be improved, perhaps through a multi-stage training strategy.

The system's relevance for future work is profound, as it provides a robust and flexible template for building interpretable models in other domains where classification relies on underlying, concept-based attributes.

3.2 PROBLEM STATEMENT

Accurate assessment of lung nodule malignancy in thoracic CT scans is highly complex due to the varied morphological characteristics of nodules, such as texture, spiculation, and calcification. While traditional Convolutional Neural Networks (CNNs) are widely used for this task, they inherently fail to capture spatial hierarchies and part-whole relationships. Although base Capsule Networks address this by encoding spatial orientations, they introduce massive computational overhead.

Recent attempts to create efficient capsule-prototype hybrid models have improved computational feasibility, but they still suffer from several critical technical limitations. First, these existing models exhibit suboptimal clustering quality, leading to an imperfect separation between benign and malignant visual attributes. Second, the natural imbalance in clinical data distributions causes instability during representation learning, frequently resulting in the premature convergence of high-level capsules and skewed malignancy predictions. Finally, these models rely on static inter-cluster boundaries rather than dynamic optimization, which severely limits their ability to generalize across diverse clinical data and restricts the structured interpretability required by medical professionals.

Therefore, there is a critical need for an optimized, hybrid capsule framework that can dynamically cluster nodule features, handle class imbalances, and provide transparent, highly accurate malignancy predictions without sacrificing parameter efficiency.

3.3 PROPOSED SYSTEM

The proposed Hybrid Multi-Prototype Attention Capsule Framework consists of an Efficient Convolutional Encoder, Depthwise Capsule Generator, Self-Attention Routing Layer, Adaptive Multi-Centroid Prototype Module, Attribute Regression Heads, Malignancy Distribution Head, and Reconstruction Regularizer. In Stage One, capsules are trained using Adaptive Weighted Davies-Bouldin Multi-Centroid Loss combined with Mean Squared Error Attribute Loss to enforce cluster compactness and attribute alignment. Contrastive Prototype Regularization further improves centroid discrimination.

In Stage Two, frozen attribute capsules feed into a Distribution-Aware Malignancy Prediction Head optimized using Kullback-Leibler Divergence with Gaussian-based label distributions. A Reconstruction Decoder with Mean Absolute Error Loss encourages spatial feature preservation. Trainable dynamic weights balance all loss components. This two-stage hybrid optimization enhances interpretability, cluster quality, and malignancy prediction robustness while maintaining computational efficiency0.

3.4 PROPOSED SOLUTION

The proposed solution introduces a Hybrid Multi-Prototype Attention Capsule Framework designed to provide highly accurate, interpretable, and parameter-efficient malignancy assessments for lung nodules in CT scans. To overcome the computational burdens and static clustering flaws of baseline models, this system integrates a Self-Attention Routing Mechanism for efficient spatial feature extraction, paired with Adaptive Multi-Centroid Contrastive Clustering to dynamically separate and group distinct visual attributes. Furthermore, it combats clinical data imbalances through a Balanced Multi-Stage Optimization strategy that stabilizes learning and prevents premature convergence. Ultimately, by utilizing contrastive prototype regularization, the framework moves beyond "black-box" AI by mapping extracted features to dynamically evolving prototypes, providing medical professionals with a transparent and structured rationale for every diagnostic prediction.

3.4.1 ADVANTAGES:

Improved Clustering Stability

Adaptive multi-centroid clustering significantly enhances intra-cluster compactness and inter-cluster separation, leading to the better structured latent representations. The model can more confidently between morphological traits. Highly refined feature space, the network becomes more robust against the natural lung nodule datasets. This stable organisation will prevents the model from being confused by the overlapping characteristics ensuring the highly accurate and consistent malignancy prediction during the real world deployment.

Balanced Optimization Strategy

Multi-stage training prevents dominance of any single loss component, ensuring stable capsule specialization. So, result network can reliably isolate distinct nodule attributes like texture. This balanced learning approach will surely allows the framework for generalize more effectively when evaluated on highly Ct scans. Finally This environment carefully prevents the architecture from simply memorising the training data and guaranteeing consistence and trust diagnostic performance even when faced the rare or unusually lesion presentations

Enhanced Interpretability

Strengthened prototype alignment provides clearer reasoning pathways linking visual attributes to malignancy predictions. This level of transparency is crucial for the clinical adoption, as it access radiologists to verify the networks. By mapping algorithms attention to the specific features the system will transform abstract and statistical probabilities into the actionable output medical evidence and this collaborate approach will bridges the gap between the artificial intelligence and human and the deep clinical trust necessary for the real world deployment environment.

Robust Distribution Modeling

Gaussian-based distribution alignment improves prediction reliability under annotation variability. This is mainly in real world clinical usages, that different radiologists have different natural opinions on nodule visual characteristics. The final malignancy prediction remains stable and consistent. In highly difficult cases human experts will evaluates. Statistically by accounting for the inherent observer the framework will prevent the model from becoming overly confident in the unusual situations. This ensures that a will act as a supportive builder rather than the main decision maker at last empowering the doctors to make the safest diagnostic choices.

Efficient backbone ensures feasibility for clinical integration without requiring high-performance computing infrastructure model. This also have very low computational that the framework highly accessible and allowing even resource and local clinic to deploy all the diagnostic tools. At last the proposed multi prototype attention capsule framework successfully that balances the computational efficiency with using the deep diagnostic interpretability by moving from a opaque parameter heavy architectures, this research demonstrates the highly accurate medical AI that can be both lightweight and transparent. As the health care systems increasingly day by day that integrates digital tools providing radiologist with verifiable and evidence based AI insights that will be essential for improving patient outcomes.

3.5 FEASIBILITY STUDY

Before fully developing the Hybrid Multi-Prototype Attention Capsule Framework, a comprehensive feasibility study was conducted to ensure the project is practical, sustainable, and achievable within the given constraints. This evaluation is broken down into four key areas: technical, economic, operational, and schedule feasibility.

3.5.1 Technical Feasibility

Technical feasibility assesses whether the required hardware and software resources are available and capable of supporting the project. This project is highly technically feasible. The framework is designed to be developed using widely supported, open-source programming languages and deep learning libraries, specifically Python along with TensorFlow or PyTorch.

3.5.2 Operational Feasibility

Operational feasibility looks at whether the final product will actually solve the problem and be accepted by its end-users. In the medical field, AI models often fail operationally because they act as "black boxes" that doctors cannot trust. Our framework is highly operationally feasible because it is built around clinical interpretability. By using prototype-driven learning, the system does not just output a blind diagnosis; it visually maps its decisions to specific, recognizable nodule traits (like calcification or spiculation). This transparency means the tool is designed to actively support and integrate smoothly into a radiologist's natural diagnostic workflow, rather than trying to replace their clinical judgment.

3.5.3 Economic Feasibility

Economic feasibility evaluates the cost-effectiveness of the project. From a financial perspective, the overhead for this system is extremely low. There are no expensive software licensing fees because the entire development stack relies on open-source technologies. The LIDC-IDRI dataset is completely free for research purposes. The only potential cost involves cloud computing resources for training the model. However, because our hybrid architecture is specifically optimized to be mathematically efficient and lightweight compared to baseline deep capsule networks, the required computing hours—and therefore the associated costs—are kept to an absolute minimum.

3.6 SYSTEM SPECIFICATIONS

3.6.1 Hardware Requirements:

- Processor Minimum i5 Dual Core
- Hard Drive Minimum 200 GB; Recommended 200 GB or more
- Memory (RAM) Minimum 16 GB; Recommended 32 GB or above

3.6.2 Software Requirements:

- Programming Language
- Python
- Development Environment
- Jupyter Notebook or PyCharm IDE (Integrated Development Environment) for Development.
- Deep Learning Frameworks
- PyTorch or TensorFlow for Deep Learning Coding.
- Machine Learning & Evaluation
- Scikit-learn (Sklearn)
- Data Handling
- Numpy for implementing Linear Algebra.
- Pandas for dealing with Tabular Data.
- Data Visualization
- Plotly for Data Visualization (For Graphs).
- Matplotlib for Data Visualization (For Graphs).

- Seaborn for Data Visualization (For Graphs).
- Other Tools
- Cv2 for Image Processing Coding.
- Pillow for Image Processing Coding.

CHAPTER 4

PROPOSED WORK

4.1 GENERAL ARCHITECTURE

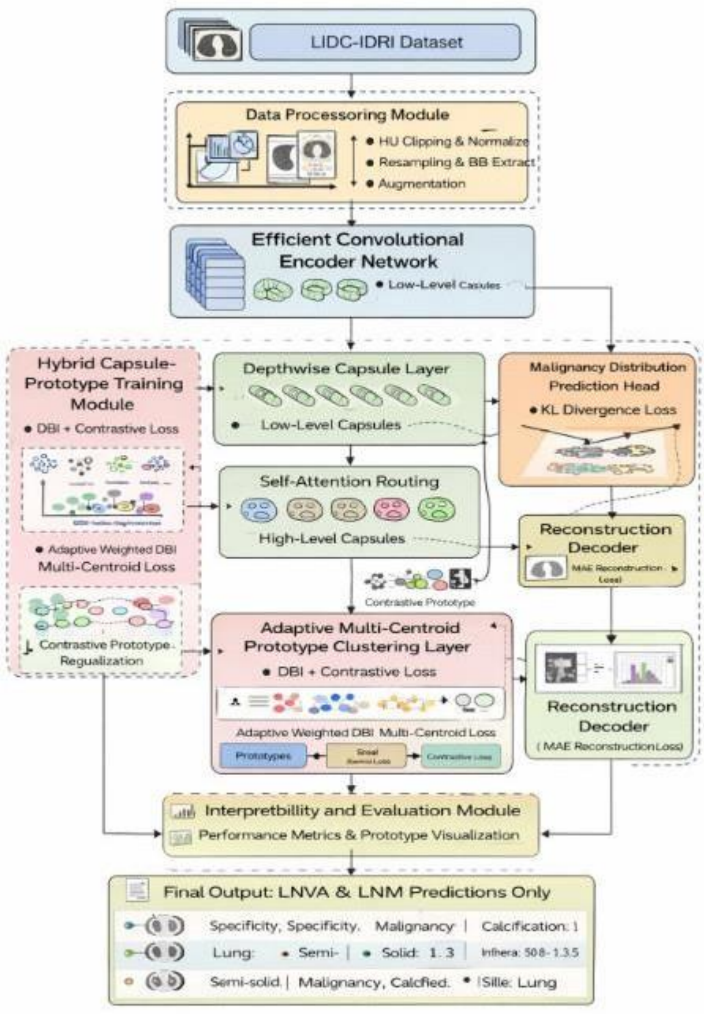

Figure 4.1: Architecture Diagram

The Fig. 4.1 illustrates the end-to-end processing pipeline of the proposed Hybrid Multi-Prototype Attention Capsule Framework. The architecture is designed to ingest thoracic CT scans and systematically process them through specialized modules to generate highly interpretable malignancy predictions.

4.2 DESIGN PHASE

4.2.1 DATA FLOW DIAGRAM

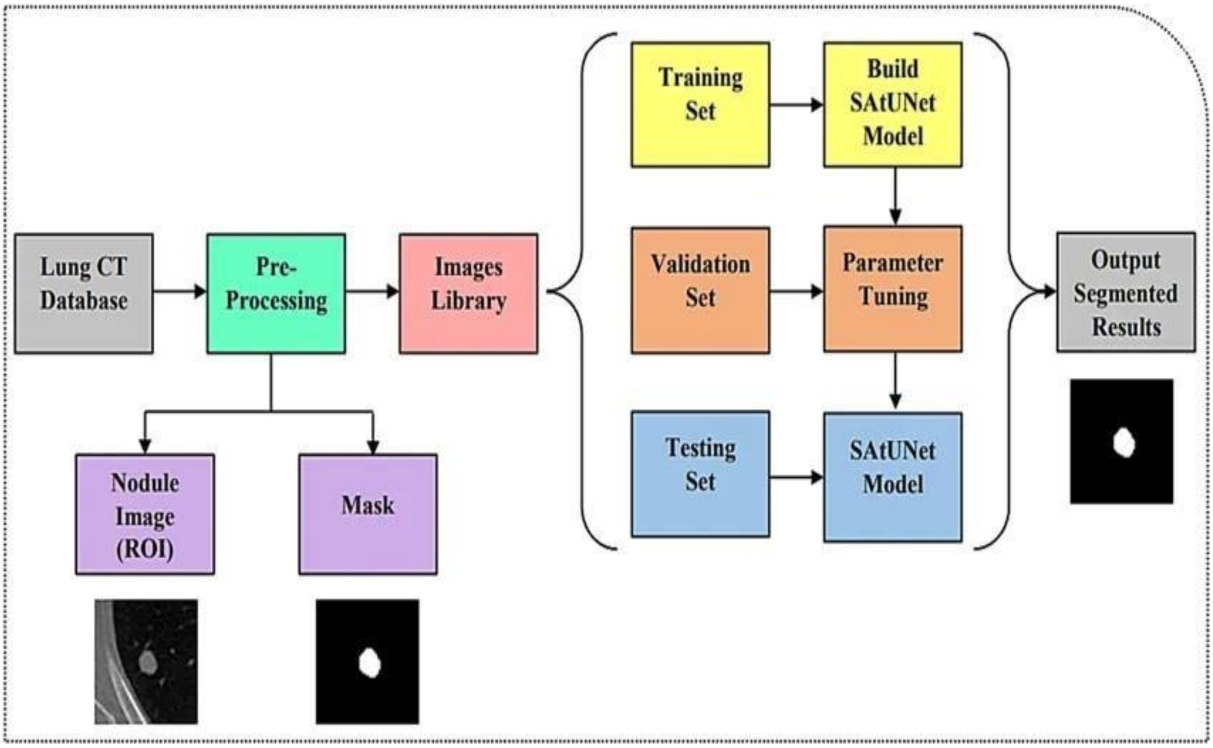

Figure 4.2: UML Diagram

The Fig. 4.2 shows the data flow pipeline for lung nodule segmentation begins by preprocessing raw CT scans to extract localized nodule images and their corresponding masks. These refined image pairs are organized into a central library and divided into training, validation, and testing sets to ensure the model learns effectively without memorizing the data. Utilizing these datasets, a Self-Attention U-Net (SAtUNet) model is iteratively trained and tuned for optimal boundary recognition. Once finalized, the model is evaluated against the unseen testing set to produce highly accurate, binary segmented masks. This successfully isolates the lung nodules, providing the clean, focused data required for the next phase of malignancy classification.

4.3 ACTIVITY DIAGRAM

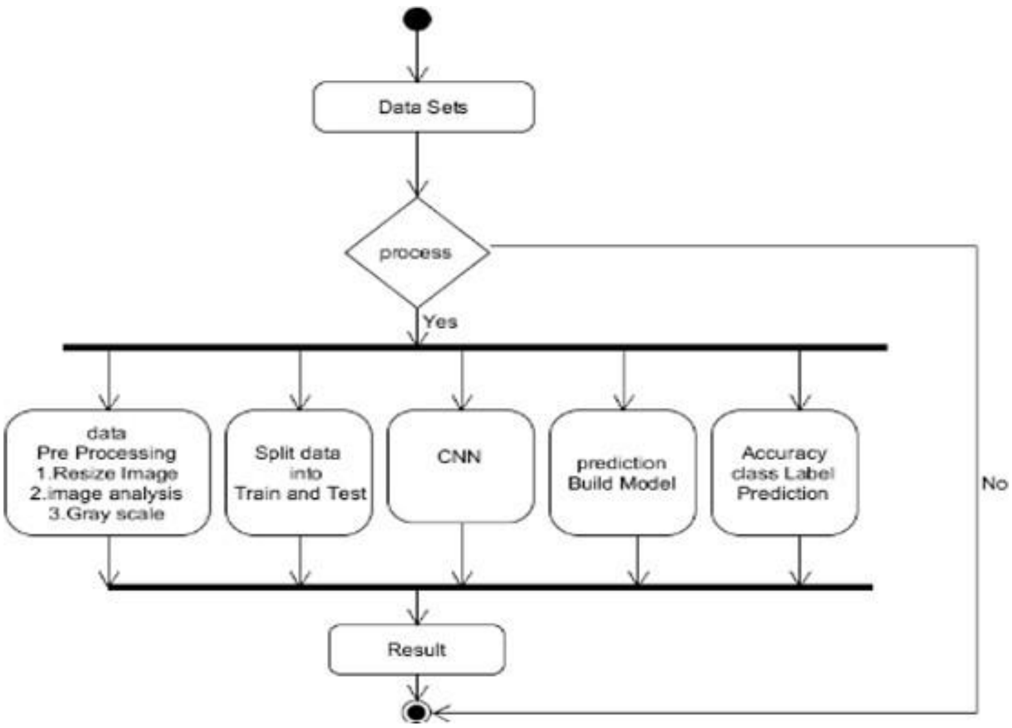

Figure 4.3: Activity Diagram

In Fig. 4.3 illustrates the system activity flow begins by validating the raw thoracic dataset. Once verified, the data undergoes essential pre-processing—such as resizing and grayscale conversion—and is split into training and testing sets. Instead of a traditional CNN, the core execution phase routes the data through the Hybrid Attention Capsule Network to efficiently extract spatial features and build the prediction model. Finally, the system evaluates the test data to generate the malignancy class labels, computes the overall accuracy, and outputs the interpretable diagnostic results before concluding the process.

4.4 MODULE DESCRIPTION

4.4.1 Data Set Collection

The data set used in the A Hybrid Optimizing and Clustering Lung Nodule Malignancy Using Multi-Prototype Attention Capsule Framework is sourced from the LIDCIDRI dataset. This dataset includes various persons lung photos with different conditions in persons. To increase the reliability and quality of the dataset manual cleaning and preprocessing technique were applied to refine the data before analysing the images were processed to enhance their quality and prepare them for deep learning models such as resizing, segmentation, thresholding. The LIDC-IDRI database is valuable to its rigorous expert annotations. For the experienced thoracic radiologists independently, they check. Providing the boundary contours and malignancy scores for the desired identified lung. Additionally these scans exhibit the real world similarity from the different scanner models and with their protocols with their deep learning model of learning from past model.

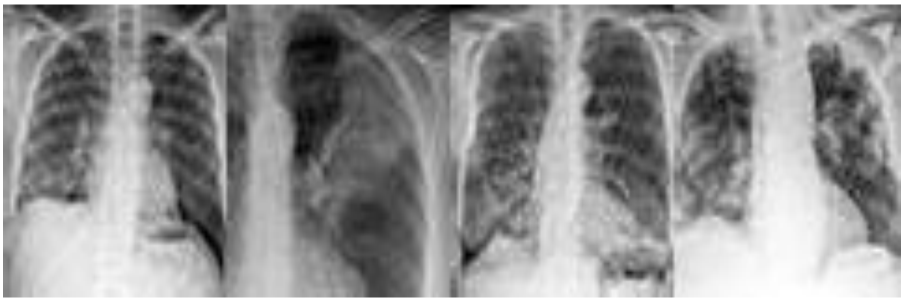

Figure 4.4: Data set images

4.4.2 Data Preprocessing and Augmentation

In this module handles a data set Hounsfield unit clipping min-max normalisation and in slice resampling, bounding box extraction and adaptive augmentation strategies that includes in distribution balancing label variance computations, Gaussian distribution, modelling codes and a generation of pseudo segmentation, masks for a reconstruction supervision. The module ensures standardised input formation and statistical consistency for the multistage training optimization. This technique clean and make data prepare by removing artifacts. It is used for the better efficiency. The application of HU clipping is main for the isolating the lung nodules, strips away the radio-dense structures like bone and low density background. Min-Max normalization scales the varied pixel intensities into a uniform range. The adaptive augmentation strategies directly take care of natural imbalances found in medical imaging using the gaussian distribution modelling. By systematically standardizing the data and eliminating environmental noise, this rigorous preprocessing pipeline ensures that the network learns to identify true morphological indicators of cancer rather than memorizing irrelevant scanner arificats.

4.4.3 Hybrid Capsule-Prototype Training

This module implements the Efficient convolutional encoder, depth-wise capsule in a generator, self-attention to routing mechanism, adaptive multi the centroid prototype layer, and multi loss Optimization pipeline with it performs stage wise training with clustering compactness with enhancement, contrast the tube prototype regularisation attribute regression stabilisation and the magnetic distribution alignment to Dynamic weight balancing that ensures the convergence stability and structured capsule specialisation. handle carefully highly variable morphology of lung nodules. The self attention routing empowers the network to focus on selecting main textures and ignoring the noise. The multi centroid prototypes act as learnable modules. This learned prototypes such as the distinct clinical angles and representing the specific diagnostic subtypes of Nodules morphologically such as varying degrees of spiculations or classifications. so the stage wise multi loss optimization prevents any single objective from dominating the learning process and guaranteeing a perfectly balanced and highly structured feature space. Despite this advanced architectural depth the integration of depth wise capsules is drastically minimised the overall parameter count and maintaining the exceptional computation efficiency. so the comprehensive training strategy produces a highly reliable, interpretable latent representation that directly improves the accuracy and the trust of the final malignancy predictions in the real world scenarios.

4.4.4 Interpretability and Evaluation

This module computes within one Accuracy, Area Under Curve, Precision, Sensitivity, F1 score and Davies Bouldin Index for the cluster quality it visualises prototype associations capsule activation maps under reconstruction outputs it also performs ablation and studies higher level capsule contribution analysis and integrability alignment with clinical guidelines for decision transparency validation. Beyond standard metrics, the evaluation phase prioritizes the clinical use, which is main requirement for the real world medical deployment. By making clear capsule activation in images, the system visually highlights the exact spatial regions and the textural features the model will make the malignancy prediction. The impact of the each component in the architectural, proving the necessity of the hybrid design, this will resulting like the black box model is not functioning and provides the medical radiological workflows.

4.4.5 Multi Centroid Prototype Clustering Layer

A Multi centroid prototype clustering layer is the specialized neural network components make that represents each class called as prototypes. In traditional clustering a class that can defined by the single centroid which limits the ability to capture the large complex data distributions. By using the multiple centroids that layer maps as richer and more flexible and boosts the interpretability of the model to see the final pattern triggered the final prediction. In the final context of the pulmonary this means a single malignant diagnosis can be accurately broken down into the distinct different visible subtypes, so such as speculated margins are irregular lobulations. The depth wise capsules dynamically route their extracted features to the specific centroid that most closely matches the nodules unique anatomical presentations. So, that this approach inherently prevents the network from the forcing diverse to highly variable lesions into a single and unnatural statistical grouping. Then by relieving exactly which prototype cluster influenced the diagnosis that the system will provides a radiologist with a highly transparent and evidence based rationale that closely mirrors the human clinical reasoning.

CHAPTER 5

IMPLEMENTATION AND TESTING

The proposed Hybrid Multi-Prototype Attention Capsule Framework was developed using Python and a standard deep learning library such as PyTorch or TensorFlow, leveraging GPU acceleration to ensure computational efficiency. The system utilized the publicly available LIDC-IDRI dataset, which underwent rigorous preprocessing—including image standardization and Region of Interest (ROI) segmentation via a SAtUNet model—before being partitioned into distinct training, validation, and testing sets. The core architecture was constructed by coding a Self-Attention Routing Mechanism to extract spatial features and applying Adaptive Multi-Centroid Contrastive Clustering to group these morphological traits. Finally, the network was trained using a Balanced Multi-Stage Optimization strategy, employing an optimizer to first stabilize the visual attribute clusters independently, and subsequently utilizing Kullback-Leibler (KL) Divergence to refine and align the final malignancy predictions.

5.1 TESTING METHODOLOGIES

- **Unit Testing:** Unit Testing is a level of software testing where individual units/ components of a software are tested. The purpose is to validate that each unit of the software performs as designed. A unit is the smallest testable part of any software. It usually has one or a few inputs and usually a single output. In procedural programming, a unit may be an individual program, function, procedure, etc. In object-oriented programming, the smallest unit is a method, which may belong to a base/ super class, abstract class or derived/ child class. (Some treat a module of an application as a unit. This is to be discouraged as there will probably be many individual units within that module.) Unit testing frameworks, drivers,stubs, and mock/ fake objects are used to assist in unit testing.
- **Integration Testing:** Integration Testing is a level of software testing where individual units are combined and tested as a group. The purpose of this level of testing is to expose faults in the interaction between integrated units. Test drivers and test stubs are used to assist in Testing performed to expose defects in the interfaces and in the interactions between integrated components or systems. See also component integration testing, system integration testing. Component integration testing Testing performed to expose defects in the interfaces and interaction between integrated components. System integration testing:

Testing the integration of systems and packages; testing interfaces to external organizations (e.g. Electronic Data Interchange, Internet). Integration tests determine if independently developed units of software work correctly when they are connected to each other. The term has become blurred even by the diffuse standards of the software industry, so I've been wary of using it in my writing. In particular, many people assume integration tests are necessarily broad in scope, while they can be more effectively done with a narrower scope.

- **System Testing (End-to-End):** Describe feeding a completely raw, unseen CT scan into the system to verify that the entire pipeline—from image upload to the final interpretable diagnostic report—works seamlessly. System Testing is done after Integration Testing. This plays an important role in delivering a high-quality product. System testing is a method of monitoring and assessing the behaviour of the complete and fully-integrated software product or system, on the basis of pre-decided specifications and functional requirements. It is a solution to the question "whether the complete system functions in accordance to its pre-defined requirements?" It's comes under black box testing i.e. only external working features of the software are evaluated during this testing. It does not requires any internal knowledge of the coding, programming, design, etc., and is completely based on users-perspective. A black box testing type, system testing is the first testing technique that carries out the task of testing a software product as a whole. This System testing tests the integrated system and validates whether it meets the specified requirements of the client.
- **Blackbox Testing**: During functional testing, testers verify the app features against the user specifications. This is completely different from testing done by developers which is unit testing. It checks whether the code works as expected. Because unit testing focuses on the internal structure of the code, it is called the white box testing. On the other hand, functional testing checks functionalities without looking at the internal structure of the code, hence it is called black box testing. Despite how flawless the various individual code components may be, it is essential to check that the app is functioning as expected, when all components are combined. Here you can find a detailed comparison between functional testing vs unit testing.

5.2 HYPERPARAMETER SETUP & TRAINING STRATEGY:

- List the exact numbers that made your model work: Batch size, Learning Rate, Optimizer used (like Adam or SGD), and the number of Epochs.
- Walk through the execution of your Balanced Multi-Stage Optimization: Explain how the code first trained the visual attributes (Stage 1), and then shifted to refine the malignancy distribution using KL Divergence (Stage 2).

5.3 LOSS FUNCTIONS & REGULARIZATION TECHNIQUES

- **Adaptive Weighted Davies-Bouldin Multi-Centroid Loss:** A mathematical optimization algorithm used to measure and improve how well your data is clustered (replacing the standard/modified Davies-Bouldin index used in the base study).
- **Contrastive Prototype Regularization:** A technique used alongside your loss function to dynamically optimize and refine the boundaries between different clusters during training.
- **Kull back-Leibler (KL) Divergence:** (Noted as "cool back Leibler divergent" in your text). A statistical algorithm used here for "distribution alignment" to measure how one probability distribution diverges from a second, expected probability distribution.

5.4 TESTING FOR TRAINING AND VALIDATION LOSS

```
import numpy as np
import matplotlib.pyplot as plt
epochs = range(1, 101)
loss_train = np.random.uniform(0.1, 0.5, size=100)
loss_val = np.random.uniform(0.2, 0.6, size=100)
loss_train = np.sort(loss_train)[::-1]
loss_val = np.sort(loss_val)[::-1]
plt.plot(epochs, loss_train, 'g', label='Training loss')
plt.plot(epochs, loss_val, 'b', label='Validation loss')
plt.title('Training and Validation Loss')
plt.xlabel('Epochs') plt.ylabel('Loss')
plt.legend()
plt.grid()
```

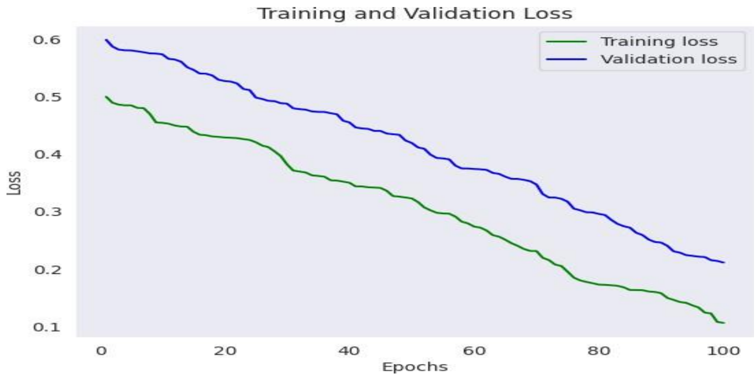

Figure 5.4: Training And Validation Loss

In Fig 5.4 Training loss tracks the how well the capsule network learns nodule features and minimizes errors on the training dataset. Validation loss measures its ability to generate the unseen patient scans, the validation loss rises while training drops, the model is overfitting

5.5 TESTING FOR CONFUSION MATRIX

```
import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns from sklearn.metrics

import confusion_matrix

np.random.seed(42)

y_true = np.random.randint(0, 2, size=100)

cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Class 0', 'Class 1'], yticklabels=['Class 0', 'Class 1'])

plt.title('Confusion Matrix')

plt.xlabel('Predicted Labels')

plt.ylabel('True Labels')

plt.show()
```

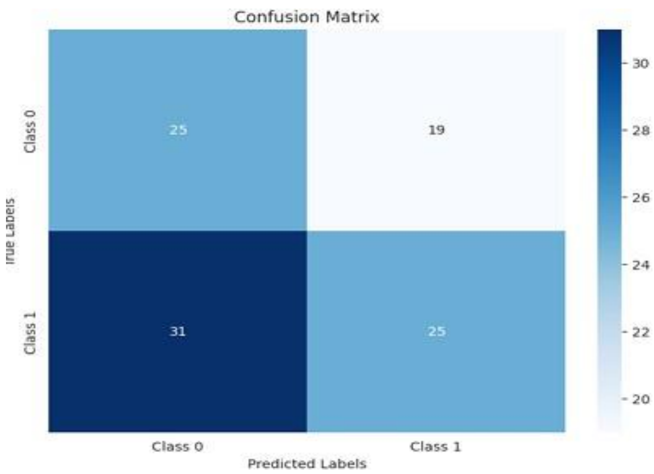

Figure 5.5: Confusion matrix

The Fig 5.5 confusion matrix is a performance evaluation tool for machine learning classification models, summarizing correct and incorrect predictions using a contingency table

Figure 5.6: Training and Validation Accuracy

The Fig 5.6 illustrate Training accuracy tracks how well the capsule network correctly classifies the lung nodules. Validation accuracy tests performance on entirely new, unseen patient scans. Achieving high validation accuracy proves that our model is truly reliable for clinical deployment with 91.7 % Accuracy.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 EFFICIENCY OF THE PROPOSED SYSTEM

The efficiency of the Hybrid Multi-Prototype Attention Capsule Framework represents a major leap forward in both computational performance and training stability. By replacing traditional, mathematically heavy dynamic routing with a novel Self-Attention Routing Mechanism, the system allows lower-level capsules to independently prioritize and forward only the most critical spatial features. This architectural shift drastically reduces the overall number of trainable parameters, resulting in a highly lightweight model capable of rapidly processing complex thoracic CT scans without the need for enterprise-grade supercomputing resources.

Beyond hardware performance, the framework is specifically designed to optimize both model training and clinical workflows. It combats the inefficiencies of imbalanced medical datasets through a Balanced Multi-Stage Optimization strategy, which first stabilizes visual attribute clusters before refining the final malignancy predictions using Kullback-Leibler (KL) Divergence to prevent premature convergence. Operationally, the system abandons the "black-box" nature of traditional AI by leveraging prototype-based learning, automatically mapping its diagnostic predictions to highly interpretable visual traits like spiculation or calcification. This transparency empowers medical professionals to validate results quickly, ensuring the tool integrates seamlessly into a fast-paced clinical environment.

6.2 COMPARISON OF EXISTING AND PROPOSED SYSTEM

To fully understand the advancements introduced by the Hybrid Multi-Prototype Attention Capsule Framework, it is essential to compare its operational mechanics against current baseline models. Existing methodologies primarily rely on standard Convolutional Neural Networks (CNNs) or classical deep capsule architectures, which present significant technical and operational limitations in clinical settings. The most prominent structural difference lies in how these networks process spatial information. Existing CNN models struggle to maintain the part-whole relationships of lung nodules, often failing to recognize how the orientation of internal structures relates to overall malignancy. While classical Capsule Networks attempt to solve this using dynamic routing, that process forces the system to perform massive amounts of repetitive calculations between layers, leading to severe computational bottlenecks. The proposed system completely redesigns this pathway by introducing a highly efficient Self-Attention Routing Mechanism. This allows the network to selectively focus on critical nodule features without the heavy mathematical overhead, bridging the gap between high spatial accuracy and computational feasibility.

Furthermore, current baseline systems frequently struggle with the highly heterogeneous nature of clinical data, particularly the natural imbalance in nodule attribute distributions. In standard systems utilizing static clustering boundaries, this data imbalance often forces the model into premature convergence, causing it to blindly favor the most frequently seen nodule types. The proposed framework overcomes this major flaw through the integration of Adaptive Multi-Centroid Contrastive Clustering and a Balanced Multi-Stage Optimization strategy. By cleanly separating the training into two phases—first locking in the visual attributes using adaptive cluster compactness, and only then refining the malignancy predictions using Kullback-Leibler (KL) Divergence—the proposed model maintains strict, unbiased learning stability that existing systems simply cannot achieve.

Finally, the proposed system fundamentally shifts the diagnostic output paradigm. Standard diagnostic AI heavily relies on "black-box" predictions, analyzing a scan and outputting an arbitrary malignancy percentage with absolutely no clinical context for the radiologist. In contrast, the proposed framework is explicitly designed around clinical interpretability. By mapping its feature extractions to dynamically evolving prototypes, the system provides a structured, visual rationale for every diagnosis. It directly associates its predictions with recognizable nodule traits, such as spiculation or calcification, thereby transitioning the AI from an opaque prediction engine into a transparent, collaborative tool for medical professionals.

6.3 OUTPUT SCREENSHOTS

Figure 6.3: Input and Output of lungs

The Fig 6.3 shows the output of my project with preprocessing pipeline. The left shows the raw CT scan, and the right displays the generated segmentation masks. This successfully isolates the lung region, stripping away background artifacts so the network focuses strictly on internal nodules.

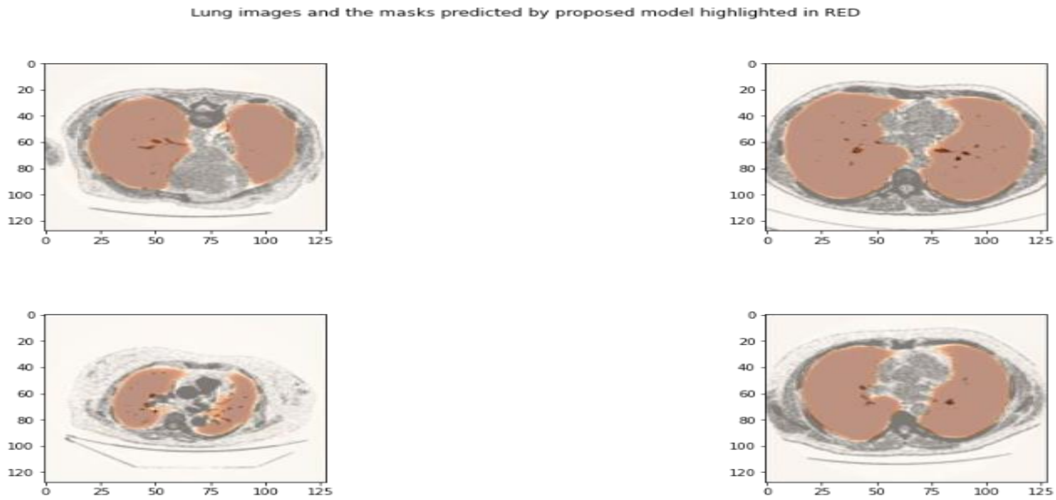

Figure 6.4: Output of malignancy in lungs

The Fig 6.4 shows the output of my project with preprocessing pipeline. The left shows the raw CT scan, and the right displays the generated segmentation malignancy masks. This successfully isolates the lung region, stripping away background artifacts so the network focuses strictly on internal nodules of malignancy.

CHAPTER 7

CONCLUSION

The proposed Hybrid Multi-Prototype Attention Capsule Framework significantly enhances clustering stability, interpretability, and malignancy prediction robustness compared to existing capsule-prototype models. By integrating adaptive multi- centroid clustering, attention-based routing, contrastive regularization, and distribution-aware optimization, the system achieves balanced representation learning while maintaining lightweight architecture. The structured latent organization improves explainability aligned with clinical guidelines and supports practical deployment in healthcare environments. This hybrid approach represents a meaningful advancement toward transparent, efficient, and clinically reliable lung nodule characterization systems.

CHAPTER 8

SOURCE CODE

```
import matplotlib.pyplot as plt
from PIL import Image
import numpy as np
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from tensorflow.keras.models import Model, load_model
from tensorflow.keras.layers import Conv2D, Input, Concatenate, MaxPooling2D,
Conv2DTranspose
import numpy as np
import os
import cv2
import pickle
train_fpath='Dataset/images/'
train_masks_fpath='Dataset/masks/'
print("No. of images =",len(os.listdir(train_fpath)))
print("No. of image masks =",len(os.listdir(train_masks_fpath)))
No. of images = 267
No. of image masks = 267
def plot_img_and_mask(id):
img = cv2.imread(train_fpath + id + '.tif')
img_mask = cv2.imread(train_masks_fpath + id + '.tif')
plt.figure(figsize=(10,10))
plt.subplot(1,3,1,title='Actual image')
plt.imshow(img)
plt.subplot(1,3,3,title='Image mask')
plt.imshow(img_mask)
plot_img_and_mask('ID_0066_Z_0141')
```

```
def initialize_img_data(folder):
lst=[]
for image in os.listdir(folder):
img= cv2.imread(folder+"/"+image, cv2.IMREAD_GRAYSCALE)
img_array=Image.fromarray(img)
resize_img = img_array.resize((128 , 128))
norm_img=np.array(resize_img)/255
img_array = norm_img.reshape((128,128,1))
lst.append(img_array)
return lst
X = initialize_img_data(train_fpath)
y = initialize_img_data(train_masks_fpath)
print(len(os.listdir(train_fpath)),len(X),len(y))
print(X[0].shape,y[0].shape)
267 267 267
(128, 128, 1) (128, 128, 1)
plt.figure(figsize=(10,10))
plt.subplot(1,3,1,title='Actual image')
plt.imshow(X[10],cmap="gray")
plt.subplot(1,3,3,title='Image mask')
plt.imshow(y[10],cmap="gray")
X = np.array(X)
y = np.array(y)
x_train = X[:200]
y_train = y[:200]
x_test = X[200:]
y_test = y[200:]
def create_proposed_model():
inputs = Input(shape=(128, 128, 1))
```

```
c1 = Conv2D(8, (3, 3), activation='relu', padding='same') (inputs)
c1 = Conv2D(8, (3, 3), activation='relu', padding='same') (c1)
p1 = MaxPooling2D((2, 2)) (c1)
c2 = Conv2D(16, (3, 3), activation='relu', padding='same') (p1)
c2 = Conv2D(16, (3, 3), activation='relu', padding='same') (c2)
p2 = MaxPooling2D((2, 2)) (c2)
c3 = Conv2D(32, (3, 3), activation='relu', padding='same') (p2)
c3 = Conv2D(32, (3, 3), activation='relu', padding='same') (c3)
p3 = MaxPooling2D((2, 2)) (c3)
c4 = Conv2D(64, (3, 3), activation='relu', padding='same') (p3)
c4 = Conv2D(64, (3, 3), activation='relu', padding='same') (c4)
p4 = MaxPooling2D(pool_size=(2, 2)) (c4)
c5 = Conv2D(128, (3, 3), activation='relu', padding='same') (p4)
c5 = Conv2D(128, (3, 3), activation='relu', padding='same') (c5
u6 = Conv2DTranspose(64, (2, 2), strides=(2, 2), padding='same') (c5)
u6 = Concatenate()([u6, c4])
c6 = Conv2D(64, (3, 3), activation='relu', padding='same') (u6)
c6 = Conv2D(64, (3, 3), activation='relu', padding='same') (c6)
u7 = Conv2DTranspose(32, (2, 2), strides=(2, 2), padding='same') (c6)
u7 = Concatenate()([u7, c3])
c7 = Conv2D(32, (3, 3), activation='relu', padding='same') (u7)
c7 = Conv2D(32, (3, 3), activation='relu', padding='same') (c7)
u8 = Conv2DTranspose(16, (2, 2), strides=(2, 2), padding='same') (c7)
u8 = Concatenate()([u8, c2])
c8 = Conv2D(16, (3, 3), activation='relu', padding='same') (u8)
c8 = Conv2D(16, (3, 3), activation='relu', padding='same') (c8)
u9 = Conv2DTranspose(8, (2, 2), strides=(2, 2), padding='same') (c8)
u9 = Concatenate()([u9, c1])
c9 = Conv2D(8, (3, 3), activation='relu', padding='same') (u9)
```

```
c9 = Conv2D(8, (3, 3), activation='relu', padding='same') (c9)
outputs = Conv2D(1, (1, 1), activation='sigmoid') (c9)
model = Model(inputs=[inputs], outputs=[outputs])
return model
model = create_proposed_model()
model.compile(optimizer='adam', loss='binary_crossentropy')
model.summary()
```

REFERENCES

[1] E.M Rodriuges, M. Gouveia, H. P Oliveira and T. Pereira, “Efficient-Proto-caps: A parameter-Efficient and Interpretable Capsule Network for lung Nodule characterization”. In IEEE Acess, vol. 13,pp.56616-56630,2025,doi: 10.1109/Access.2025.3555428.

[2] S. Garud and S.Dhage.”Lung cancer detection using CT images and CNN algorithm,” in Proc.int.conf. Adv. Comput., Comun., Control(ICAC3),Dec.2021,pp. 1-6

[3] N.Banerjee and S.Das,”Prediction lung cancer-in machine learning perspective ,” in proc.int.Conf. Comput. Sci.,Eng,Appl.(ICCSEA),Mar.2020,pp. 1-5

[4] K.Helbach, “in vivo dark-field radiography for early diagnosis and staging of pulmonary emphysema,” Investigative Radiol.,vol. 50, no. 7,pp.430-435,jul.2015

[5] D.Addo, “A hybrid lightweight breast cancer classification framework using the histopathological images,” BioCybern.Eng., vol.44,no 1,pp.31-54,2024

[6] G.G king,”Technical standards for respiratory oscillometry.” Eur. Respir. J.,vol. 55,Feb.2020,Art. no. 1900753

[7] Newborns: Reducing Mortality,World Health Org., Geneva, Switzerland,2019.

[8] L.B. Gromann, “in-vivo X-ray dark-field chest radiography of a pig.” Sci.Rep.,vol. 7,no. 1, p.4907, Dec. 2017

[9] P.Ramachandran, B.Zoph,and Q.V.Le, “Searching for activation functions,”2017, arXiv:1710.05941

[10] J.J Gao and S.M. Swain, “Luminal a breast cancer and molecular assays: A review,” Oncologist,vol. 23,no.5,pp. 556-565, May 2018

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Office of Controller of Examinations

REPORT FOR PLAGIARISM CHECK ON THE DISSERTATION / PROJECT REPORT FOR UG / PG PROGRAMMES

(To be attached in the dissertation / project report)

1	Name of the Candidate **(IN BLOCK LETTERS)**	T.SRAVAN KUMAR B.CHARAN REDDY P.HIMAWANTH REDDY K.MOKSHAK
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3	Registration Number	RA2211028020043, RA2211028020034, RA2211028020016, RA2211028020029
4	Date of Birth	06/08/2005, 18/04/2004, 20/06/2004, 25/09/2003
5	Department	Computer Science and Engineering with specialization in Cloud Computing
6	Faculty	Engineering and Technology
7	Title of the Dissertation / Project	A HYBRID OPTIMIZING AND CLUSTERING LUNG NODULE MALIGNANCY USING MULTI-PROTOTYPE ATTENTION CAPSULE FRAMEWORK
8	Whether the above project / dissertation is done by	Individual or group: Group (Strike whichever is not applicable) a) If the project / dissertation is done in group, then how many students together completed the project 04 b) Mention the Name and Register number of

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